1_Class_Content

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Subject - Computer Networks

1. What is a Network?

Interview Perspective: Don't just say "connected computers." Frame it as a system designed for resource sharing and communication.

- **Definition:** A network is a collection of autonomous computing devices (nodes) interconnected by communication channels (wired or wireless) to share resources, data, and services.
- **Key Components:**
  - **Nodes:** The devices (PC, Server, Router).
  - **Links:** The medium (Fiber, Wi-Fi, Ethernet).
  - **Protocols:** The rules governing data exchange (TCP/IP, HTTP).
- **The "Why":** In an enterprise context like AT&T, networks are categorized by scale: **LAN** (Local Area Network), **WAN** (Wide Area Network), and **MAN** (Metropolitan Area Network).

2. What is an IP Address?

Interview Perspective: Focus on its role in the **Network Layer (Layer 3)** and the difference between its two main versions.

- **Definition:** An Internet Protocol (IP) address is a **logical identifier** assigned to each device on a network. It serves two primary functions: host identification and location addressing.
- **Types:**
  - **IPv4:** A 32-bit address usually written in dotted-decimal format (e.g., `192.168.1.1`). It provides roughly 4.3 billion unique addresses.
  - **IPv6:** A 128-bit address written in hexadecimal (e.g., `2001:0db8:85a3:0000:0000:8a2e:0370:7334`). Created to solve the address exhaustion of IPv4.
- **Nature:** It is **dynamic** or **static**. A device's IP changes when it moves to a different network (unlike its MAC address).

3. What is a MAC Address?

Interview Perspective: Emphasize that this is the "Physical" address tied to the hardware at the **Data Link Layer (Layer 2)**.

- **Definition:** A Media Access Control (MAC) address is a **unique, permanent hardware identifier** burned into the Network Interface Card (NIC) by the manufacturer.
- **Format:** It is a 48-bit address, represented as 12 **hexadecimal** digits, typically separated by colons or hyphens (e.g., `00:1A:2B:3C:4D:5E`).
  - The first 24 bits represent the **OUI** (Organizationally Unique Identifier) — basically the "Who made this?" (e.g., Intel, Apple).
  - The last 24 bits are the specific device serial number.
- **Key Distinction:** While an IP address gets a packet to the right *network*, the MAC address is used to deliver the frame to the specific *device* on that local segment.

Comparison Table for Quick Revision

Feature	IP Address	MAC Address
Layer	Network Layer (Layer 3)	Data Link Layer (Layer 2)
Type	Logical / Virtual	Physical / Hardware
Assignment	DHCP or Manual Configuration	Hardcoded by Manufacturer
Scope	Used for global/inter-network routing	Used for local/intra-network delivery
Format	Dotted Decimal (IPv4)	Hexadecimal

4. Why the OSI Model Was Introduced (The Need)

Interview Perspective: Do not just say "to understand networks." Frame it as a solution to a massive business and engineering problem: **vendor lock-in and incompatibility.**

The Problem (Before OSI)

In the 1970s and early 1980s, computer networking was proprietary.

- Companies like IBM, DEC, and Honeywell built their own networking architectures.
- An IBM computer could only talk to another IBM computer. It used entirely different hardware, cabling, and software protocols than a DEC computer.
- If a company wanted to connect them, it was nearly impossible or required incredibly expensive, custom-built translators.

The Solution: Why OSI was Created

The **ISO (International Organization for Standardization)** introduced the **OSI (Open Systems Interconnect) model** in 1984 to:

- **Create Open Standards:** It established an open framework so different vendors could create hardware and software that could seamlessly talk to each other (**Interoperability**).
- **Modularize Network Development:** By breaking the complex process of communication into 7 distinct, manageable layers, engineers could work on one layer without worrying about the others.
- **Standardize Interfaces:** It allowed a vendor to develop a new technology at one layer (like a new Wi-Fi standard at Layer 1) without needing to rewrite the web browser code (Layer 7).

> **Interview Pitch:** *"The OSI model was introduced to move the industry away from closed, proprietary ecosystems to an open market, allowing hardware and software from completely different manufacturers to interoperate seamlessly."*

5. What are the OSI Layers and the Use of Each Layer?

Interview Perspective: You must know the 7 layers in order. A classic mnemonic to remember them from top to bottom (Layer 7 to Layer 1) is: **A**ll **P**eople **S**eem **T**o **N**eed **D**ata **P**rocessing.

When describing each layer, always state its **Data Unit** and its **Primary Responsibility**.

```
[ Layer 7: Application ]      <- Data
[ Layer 6: Presentation ]     <- Data
[ Layer 5: Session ]          <- Data
[ Layer 4: Transport ]        <- Segments
[ Layer 3: Network ]          <- Packets
[ Layer 2: Data Link ]        <- Frames
[ Layer 1: Physical ]         <- Bits
```

Layer 7: Application Layer

- **Data Unit:** Data
- **Purpose:** This is the closest layer to the end-user. It provides network services directly to software applications (like web browsers or email clients). It does *not* mean the browser itself is at Layer 7, but the protocol the browser uses resides here.
- **Key Functions:** Service advertisement, resource identification.
- **Protocols:** HTTP, HTTPS, FTP, SMTP, DNS.

Layer 6: Presentation Layer

- **Data Unit:** Data
- **Purpose:** Acts as the translator for the network. It ensures that data sent from the application layer of one system can be read by the application layer of another.

- **Key Functions: Translation** (e.g., ASCII to EBCDIC), **Encryption/Decryption** (SSL/TLS), and **Compression** (reducing file size for transmission).

Layer 5: Session Layer

- **Data Unit:** Data
- **Purpose:** Manages the dialogues (sessions) between computers. It establishes, maintains, and terminates connections between local and remote applications.
- **Key Functions:** Authentication, authorization, and checkpointing (adding recovery points so a file transfer can resume if interrupted).
- **Protocols:** NetBIOS, RPC, PPTP.

Layer 4: Transport Layer

- **Data Unit:** Segments (TCP) / Datagrams (UDP)
- **Purpose:** Responsible for **end-to-end communication** and error-free delivery of data from the source host to the destination host.
- **Key Functions:** * **Segmentation & Reassembly:** Chopping big data into small pieces and putting them back together.
  - **Flow Control:** Preventing a fast sender from overwhelming a slow receiver.
  - **Error Control:** Ensuring data arrives without corruption (retransmitting if lost).
- **Protocols:** TCP (Reliable), UDP (Fast/Unreliable).

Layer 3: Network Layer

- **Data Unit:** Packets
- **Purpose:** Responsible for **path determination** and moving packets from the source network to the destination network across multiple links. This is where **logical addressing (IP)** lives.
- **Key Functions:** Routing (choosing the best path), Logical Addressing.
- **Devices/Protocols:** Routers, IP (IPv4/IPv6), ICMP, OSPF.

Layer 2: Data Link Layer

- **Data Unit:** Frames
- **Purpose:** Responsible for **hop-to-hop** or node-to-node delivery of data within the *same* local network. It deals with physical hardware addressing.
- **Key Functions: Physical Addressing (MAC addresses)**, Framing, Media Access Control (determining who speaks on the wire).
- **Devices/Protocols:** Switches, Bridges, Ethernet, Wi-Fi (802.11), ARP.

Layer 1: Physical Layer

- **Data Unit:** Bits (1s and 0s)
- **Purpose:** Responsible for the actual physical transmission of raw, unstructured data bits over a physical medium (cables, radio waves, light).
- **Key Functions:** Defining electrical voltages, pin layouts, cable types, data rates, and signal modulation.
- **Devices:** Hubs, Repeaters, Cables (Fiber, Coaxial, Cat6), Modems.

6. What are Segments, Packets, and Frames?

Interview Perspective: These are collectively known as **PDUs (Protocol Data Units)**. A PDU is simply data wrapped in specific header control information at each layer of the OSI model. Explain them as a **nested Russian nesting doll**—each layer adds its own envelope around the original data.

```
[Application Data]
       ↓
[Transport Header] + [Data]                      = SEGMENT (Layer 4)
       ↓
[Network Header]   + [Transport Header + Data]    = PACKET  (Layer 3)
       ↓
[Data Link Header] + [Network Header + ... ] + [Trailer] = FRAME   (Layer 2)
```

1. Segment (Layer 4 - Transport Layer)

- **What it is:** When the Application Layer passes down a massive chunk of data (like a 50MB image), the Transport layer breaks it into smaller, manageable chunks.
- **The Header Includes:** Source and Destination **Port Numbers** (to know which app sent/receives it) and **Sequence Numbers** (to put them back in the right order).
- *Note:* In UDP, this PDU is technically called a **Datagram**, while in TCP, it is called a **Segment**.

2. Packet (Layer 3 - Network Layer)

- **What it is:** The Network Layer takes the Transport Segment and wraps it in a routing envelope.
- **The Header Includes:** Source and Destination **IP Addresses**.
- **Interview Context:** Routers read packets. A packet contains all the necessary routing information to travel across multiple networks globally.

3. Frame (Layer 2 - Data Link Layer)

- **What it is:** The Data Link Layer takes the Network Packet and encapsulates it with a local delivery envelope. It adds both a **Header** and a **Trailer**.
- **The Header/Trailer Includes:** Source and Destination **MAC Addresses**, and a **CRC (Cyclic Redundancy Check)** in the trailer to detect if any bits were corrupted during physical transmission.
- **Interview Context:** Switches read frames. Frames only care about moving data from one physical device to the next adjacent device in the local network.

7. TCP, UDP, and QUIC

Interview Perspective: For a telecom/network company like AT&T, understanding these transport protocols is non-negotiable. They will want to see if you understand the trade-offs between **reliability** and **speed**, as well as modern evolutions like QUIC.

1. TCP (Transmission Control Protocol)

- **Core Nature:** Connection-oriented, reliable, and guarantees ordered delivery.
- **How it works:** It establishes a connection using a **3-Way Handshake** (SYN $\rightarrow$ SYN-ACK $\rightarrow$ ACK) before sending any data.

What is Windowing in TCP?

- **Definition:** Windowing is a **flow-control mechanism** used by TCP to manage the rate of data transmission between a sender and a receiver.
- **The Problem:** If a fast server sends data too quickly, it will overflow the receiver's memory buffer, causing dropped packets.
- **The Solution (Sliding Window):** The receiver specifies a **Window Size** in its TCP header. This number tells the sender, *"You can send me exactly X bytes of data before you must stop and wait for an Acknowledgment (ACK) from me."*
- **Dynamic Adjustment:** If the receiver's buffer fills up, it shrinks the window size to slow the sender down. If it empties, it expands the window size to speed up data transfer.

2. UDP (User Datagram Protocol)

- **Core Nature:** Connectionless, unreliable, and unordered.
- **How it works:** It does not establish a connection. It just shoots packets (datagrams) at the destination without checking if the receiver is ready or if the data actually arrived ("Fire and Forget").
- **Why use it?** It has almost zero overhead, making it incredibly fast.
- **Use Cases:** Live video streaming, VoIP calling, online gaming, and DNS lookups—places where losing a single frame of audio/video is better than waiting for a retransmission.

3. QUIC (Quick UDP Internet Connections)

- **Core Nature:** A modern transport protocol designed by Google that combines the **speed of UDP** with the **reliability of TCP**. It forms the foundation of **HTTP/3**.

Why was QUIC needed? (The Problem with TCP)

When connecting to a secure website via TCP, you suffer from high latency due to multiple handshakes:

1. TCP Handshake (1 RTT - Round Trip Time)
2. TLS/Encryption Handshake (1-2 RTTs)

   This creates a noticeable delay before any actual web data is sent.

How QUIC Fixes It:

- **Runs on top of UDP:** QUIC bypasses the slow TCP stack entirely by running on UDP.
- **0-RTT Handshake:** QUIC combines the connection handshake and the security (TLS 1.3) handshake into **one single step**. If you have connected to the server before, it can send data instantly (**0-RTT**).
- **Solves Head-of-Line Blocking:** In TCP, if packet #1 is lost, packets #2 and #3 must wait in the buffer until packet #1 is retransmitted. QUIC allows multiple independent streams of data; if one stream loses a packet, the other streams keep moving without stopping.
- **Connection Migration:** If you switch from Wi-Fi to cellular data on your phone, a TCP connection drops and must reconnect. QUIC uses a unique "Connection ID" instead of an IP address, so your download continues seamlessly without interrupting.

Comparison Summary for the Interview

Feature	TCP	UDP	QUIC
Connection	Connection-oriented	Connectionless	Connection-oriented (Logical)
Reliability	High (Retransmissions)	None	High (Handles loss internally)
Speed/Latency	Slower (Handshake overhead)	Extremely Fast	Fast (0-RTT Handshake)
Underlying Protocol	Built directly on IP	Built directly on IP	**Built on top of UDP**
Primary Use Case	Web Browsing (HTTP/1.1, HTTP/2), Email, File Transfer	DNS, VoIP, Video Streaming	Modern Web (HTTP/3),

Feature	TCP	UDP	QUIC
			YouTube

8. What is the Three-Way Handshake?

Interview Perspective: This is the process TCP uses to establish a **reliable** connection. The goal isn't just to say "hello," but to **synchronize sequence numbers** so both sides can track which data has been sent and received.

The Steps (SYN → SYN-ACK → ACK)

1.SYN (Synchronize): Client to Server.

The client sends a packet with the **SYN flag** set and an Initial Sequence Number (**ISN**), say `X`.

- **Meaning:** "I want to start a connection with you. Here is my starting sequence number."

2.SYN-ACK (Synchronize-Acknowledge): Server to Client.

The server responds with the **SYN** and **ACK** flags set. It sends its own ISN, say `Y`, and an Acknowledgment Number of `X+1`.

- **Meaning:** "I received your request (ACK X+1). I also want to connect; here is my starting sequence number (Y)."

3.ACK (Acknowledge): Client to Server.

The client sends a final packet with the **ACK flag** set and an Acknowledgment Number of `Y+1`.

- **Meaning:** "I received your sequence number (ACK Y+1). We are now connected."

> **Why the +1?** In an interview, mention that the acknowledgment number is always the **next expected sequence number**. By sending `X+1`, the server is saying "I have everything up to `X`, send me `X+1` next."

9. Port Numbers of Main Servers

Interview Perspective: You should memorize the "Well-Known Ports" (0–1023). These are standardized by the IANA so that clients know exactly where to find specific services on a server.

Essential Ports for Freshers

Port	Protocol	Service	Use Case
20/21	FTP	File Transfer	20 is for Data; 21 is for Control (Commands)
22	**SSH**	Secure Shell	Secure remote login and file transfer (SFTP)
23	Telnet	Terminal Emulation	Unencrypted remote login (rarely used now)
25	**SMTP**	Email	Sending/routing email between servers
53	**DNS**	Domain Name System	Resolving domain names to IP addresses
80	**HTTP**	Web	Standard unencrypted web traffic
443	**HTTPS**	Secure Web	Web traffic encrypted via TLS/SSL
110	POP3	Email	Retrieving email (removes from server)
143	IMAP	Email	Retrieving email (stays on server/syncs)
161	SNMP	Management	Monitoring and managing network devices

Common Database/App Ports (Extra Credit)

- **3306:** MySQL
- **5432:** PostgreSQL
- **8080:** Often used for proxy servers or "Alternative HTTP" for dev environments.

Pro-Tip: The "Socket" Concept

If an interviewer asks what a **Socket** is, give them this formula:

Socket = IP Address + Port Number + Protocol (TCP/UDP)

- The IP address gets the data to the correct **building** (the device).
- The Port Number gets the data to the correct **apartment** (the specific application).

10. The Network Device Ecosystem

1. Hub

- **OSI Layer:** Layer 1 (Physical)
- **What it does:** A legacy device used to connect multiple computers in a local network. It has no "intelligence."
- **How it works:** When it receives a signal on one port, it blindly broadcasts (copies) that signal to **all other ports**, regardless of who the intended recipient is.
- **Key Issues:** It creates a single **Collision Domain** and security risks, as every device sees everyone else's traffic. It operates in **Half-Duplex** (devices can't send and receive at the

same time).

2. Repeater

- **OSI Layer:** Layer 1 (Physical)
- **What it does:** It amplifies or regenerates weak electrical, light, or radio signals.
- **Why it's needed:** As signals travel down a cable (like an Ethernet or Fiber cable), they suffer from **attenuation** (they get weaker over distance). A repeater boosts the signal so it can travel further without corruption.

3. Switch

- **OSI Layer:** Layer 2 (Data Link)
- **What it does:** An intelligent device used to connect devices within the *same* local network (LAN).
- **How it works:** Unlike a hub, a switch maintains a **MAC Address Table**. It looks at the destination MAC address of an incoming frame and intelligently forwards it *only* to the specific port where that device is connected.
- **Key Advantage:** Every single port on a switch is its own collision domain, allowing for **Full-Duplex** communication (sending and receiving simultaneously).

4. Bridge

- **OSI Layer:** Layer 2 (Data Link)
- **What it does:** A legacy device used to connect two separate LAN segments together to form a larger single network.
- **How it works:** It filters traffic by reading MAC addresses. If a packet is meant for a device on the same segment, it blocks it from crossing; if it's meant for the other segment, it passes it through.
- *Note for Interview:* **A modern Switch is essentially a multi-port Bridge.** Bridges usually had only 2 to 4 ports and did switching via software, whereas switches have many ports and do it via dedicated hardware chips (ASICs).

5. Router

- **OSI Layer:** Layer 3 (Network)
- **What it does:** An intelligent device used to connect **different** networks together (e.g., connecting your home LAN to the global Internet).
- **How it works:** It reads the destination **IP addresses** of incoming packets and uses a **Routing Table** alongside routing protocols (like OSPF or BGP) to determine the absolute best path for the packet to reach its destination network.

- **Key Advantage:** Routers completely block Layer 2 broadcast traffic, creating separate **Broadcast Domains**.

Other Important Devices (High-Yield for AT&T)

6. Gateway

- **OSI Layer:** Operates across all layers (Layers 4 to 7 primarily).
- **What it does:** Acts as a protocol translator. It connects two entirely different network architectures that use different protocols (e.g., connecting a legacy telecom network to an IP-based internet network).
- *Note:* Do not confuse this with a "Default Gateway," which is just the IP address of your local router.

7. Firewall

- **OSI Layer:** Layers 3, 4, and 7 (Application)
- **What it does:** A security device that monitors and filters incoming and outgoing network traffic based on a defined set of security rules.
- **Types:**
  - **Packet-Filtering (Stateless):** Looks only at source/destination IP and port numbers.
  - **Stateful Inspection:** Tracks the state of active connections (knows if a packet is part of an existing conversation).
  - **Next-Gen Firewall (NGFW):** Looks deep into the actual application payload (Layer 7) to detect malware.

8. Access Point (WAP)

- **OSI Layer:** Layer 2 (Data Link)
- **What it does:** A device that allows wireless devices (like your phone or laptop) to connect to a wired network using Wi-Fi standards. It bridges the wireless media (radio waves) to the wired media (Ethernet).

Quick Summary Reference for the Interview

Device	OSI Layer	Key Identifier Used	Primary Function
Repeater	Layer 1	None (Electrical bits)	Regenerate/Amplify signals
Hub	Layer 1	None (Electrical bits)	Blindly broadcast data to all ports

Device	OSI Layer	Key Identifier Used	Primary Function
Bridge	Layer 2	MAC Address	Connect 2 local network segments
Switch	Layer 2	MAC Address	Intelligently direct traffic within a LAN
Router	Layer 3	IP Address	Route data packets *between* different networks
Firewall	Layer 3/4/7	IPs, Ports, Payloads	Secure the network by filtering traffic

> **Classic Interview Scenario:** > *Interviewer:* "What is the difference between a Router and a Switch?"
>
> *Your Answer:* "A **Switch** connects devices to form a local network (LAN) and routes traffic within that network using **MAC addresses** at Layer 2. A **Router** connects entirely separate networks together and routes traffic between them using **IP addresses** at Layer 3."

11. What is DHCP?

Interview Perspective: Frame DHCP not just as "something that gives IPs," but as an **automation protocol** that prevents IP address conflicts and reduces network administration overhead.

1. Definition

- **DHCP** stands for **Dynamic Host Configuration Protocol**.
- It is an **Application Layer (Layer 7)** protocol that automatically assigns IP addresses and other network configuration parameters to devices joining a network.
- It operates using a **Client-Server model** and relies on **UDP** as its transport protocol (**Port 67 for the Server, Port 68 for the Client**).

2. The Problem It Solves

Without DHCP, network administrators would have to manually type an IP address, Subnet Mask, Gateway, and DNS server into every single device.

- **The Risks:** If two people accidentally type the same IP address, a **Duplicate IP Conflict** occurs, knocking both devices offline.
- **The Scale:** Imagine a telecom network like AT&T manually configuring millions of mobile devices moving across cellular towers. It would be impossible.

How It Works: The DORA Process

When an interviewer asks *"How does DHCP work?"*, the word they are looking for is **DORA**. This is a 4-step handshake process:

```
[ Client ]                                      [ DHCP Server ]
    |                                                  |
    | ─── 1. DISCOVER (Broadcast: "Anyone there?") ──>|
    |                                                  |
    | <── 2. OFFER    (Unicast: "Here is an IP") ─────|
    |                                                  |
    | ─── 3. REQUEST  (Broadcast: "I'll take it!") ──>|
    |                                                  |
    | <── 4. ACK      (Unicast: "Great, it's yours") ─|
    |                                                  |
```

1. Discover (Client → Server)

The client machine boots up and has no IP address. It sends a **DHCPDISCOVER** message out onto the local network.

- **Crucial Detail:** Because it doesn't know where the server is, this message is sent as a **Broadcast** (Destination IP: `255.255.255.255`, Destination MAC: `FF:FF:FF:FF:FF:FF`).

2. Offer (Server → Client)

Any DHCP server on the local network hears the broadcast, checks its pool of available IP addresses, reserves one, and sends a **DHCPOFFER** back to the client.

- This offer includes an available IP address, Subnet Mask, Lease Time, and the IP of the default gateway.

3. Request (Client → Server)

The client receives the offer. If there are multiple servers, it usually accepts the first offer it gets. It responds with a **DHCPREQUEST** message.

- **Crucial Detail:** This is **still a broadcast**. It tells the server that offered the IP, *"Yes, I accept your offer,"* while simultaneously letting other DHCP servers know they can release their reserved IPs back into their pools.

4. Acknowledge (Server → Client)

The server receives the request and sends a final **DHCPACK** (Acknowledgment) packet to the client, confirming that the IP address has officially been leased to that device. The client can now use the network.

3. Important Concepts to Know

What is a DHCP Lease?

DHCP addresses are not given permanently; they are **leased** for a specific period (e.g., 24 hours).

- When **50% of the lease time expires (T1 timer)**, the client will automatically attempt to renew the lease with the original DHCP server.
- If that fails, at **87.5% of the lease time (T2 timer)**, it will broadcast to find *any* available DHCP server to renew its lease.

What is APIPA (Automatic Private IP Addressing)?

If a client machine sends out a DHCPDISCOVER packet but **never receives a reply** (perhaps the server is down or the cable is unplugged), Windows/macOS devices will automatically assign themselves an APIPA address.

- **The Range:** Always starts with `169.254.X.X`.
- **Interview Tip:** If an interviewer asks, *"A user can communicate with computers in their immediate office but cannot access the Internet or external servers. Their IP is 169.254.4.5. What is the issue?"* * **Your Answer:** "The machine is using an APIPA address, meaning it failed to reach the DHCP server. It can only communicate locally on its immediate physical switch segment, but cannot route out to the internet."

What Parameters Does DHCP Provide?

It gives the client 4 key things:

1. **IP Address:** The unique identity on the network.
2. **Subnet Mask:** Defines which part of the IP is the network id and which part is the host id.
3. **Default Gateway:** The IP address of the local router (the way out of the network).
4. **DNS Server IP:** The address of the server that will translate domain names (like google.com) into IP addresses.